BootIT

Ai Ting Lee

Based on slides by Niek Janssen
-- Based on slides by Sebastian Mateos Nicolajsen
---- Based on slides by Claus Brabrand

Welcome!!!

Teachers

Ai Ting Lee

- She/Her
- From Taiwan 🧋 🧋 🧋
- Mandarin (Native), English, Danish (Terrible)
- Full-stack developer at Dreamdata
  ITU TA in Practical Concurrent and Parallel Programming
- MSc in Software Design (2024 - June 2026)
- Bachelor of Education
- Scuba diver

Amanda Lima Soares Da Cunha

- She/Her
- From Brasil
- Languages: English 🇬🇧 , Portuguese 🇧🇷 , when drunk Spanish, lidt dansk 🇩🇰
- Software Engineer at Sealytix; TA at ITU in Intro to AI, Algorithms & Study Lab
- MSc Software Design (2024 - N/A)
- BSc Architecture
- Yoga & reading loads of books (always accept recommendations)

Avery Elise Brunsman

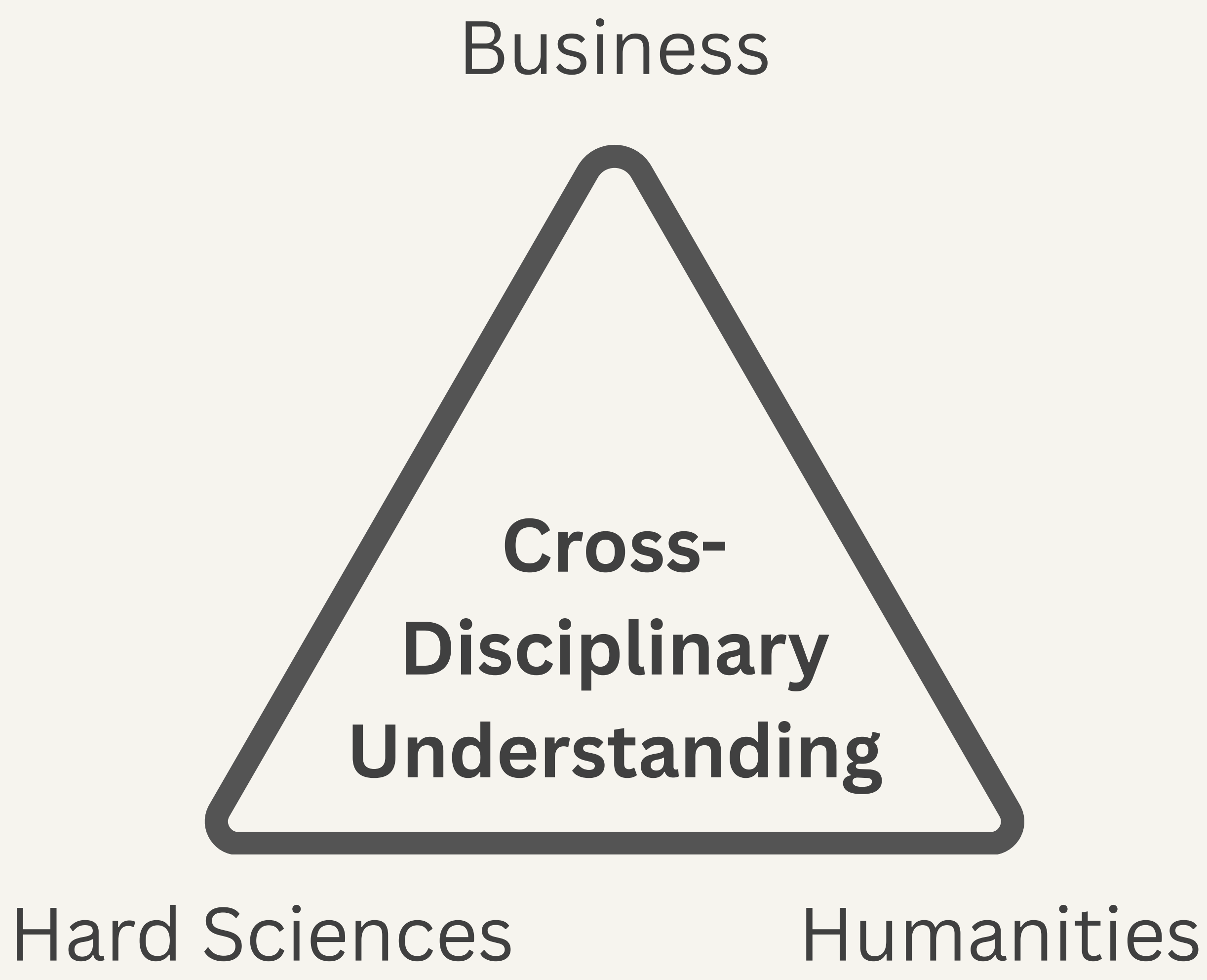
- She/her
- From San Francisco, California
- Second year SD
- Bachelor in Graphic Communication
- Student job in graphic design
- Volunteer barista at Analog
- Love hiking and sauna

Johannes Alexander Hackl

- He/Him
- From Austria 🇦🇹
- Languages: English 🇬🇧, German 🇦🇹, Portuguese 🇧🇷, very bad Danish 🇩🇰
- Software Developer at Calibras;
  TA at ITU in Discrete Mathematics, Algorithms & Study Lab
- MSc Software Design (2024 - N/A)
- BSc Architecture 🏛️
- Footvolley 🏐 & Bouldering 🧗

IT University of Copenhagen

ITU Mission: **"... Create Value with IT in Denmark"**



MSc in Software Design (SD)

1st semester - Autumn

Introductory Programming 15 ECTS	Discrete Mathematics 7.5 ECTS	Software Engineering 7.5 ECTS

2nd semester - Spring

Introduction to Database Systems 7.5 ECTS	Algorithms and Data Structures 7.5 ECTS	Specialisation Course 7.5 ECTS	Elective or Specialisation course 7.5 ECTS

3rd semester - Autumn

Research Project 7.5 ECTS	Specialisation Course 7.5 ECTS	Specialisation Course 7.5 ECTS	Elective or Specialisation course 7.5 ECTS

4th semester - Spring

Thesis 30 ECTS

MSc in Software Design (SD)

1st semester - Autumn

Introd… 15 EC…

Dis… 7.5…

2nd sem…

Introd… 7.5 E…

…tructures

Specialisation Course 7.5 ECTS

Elective or Specialisation course 7.5 ECTS

3rd semester - Autumn

Research Project 7.5 ECTS

Specialisation Course 7.5 ECTS

4th semester - Spring

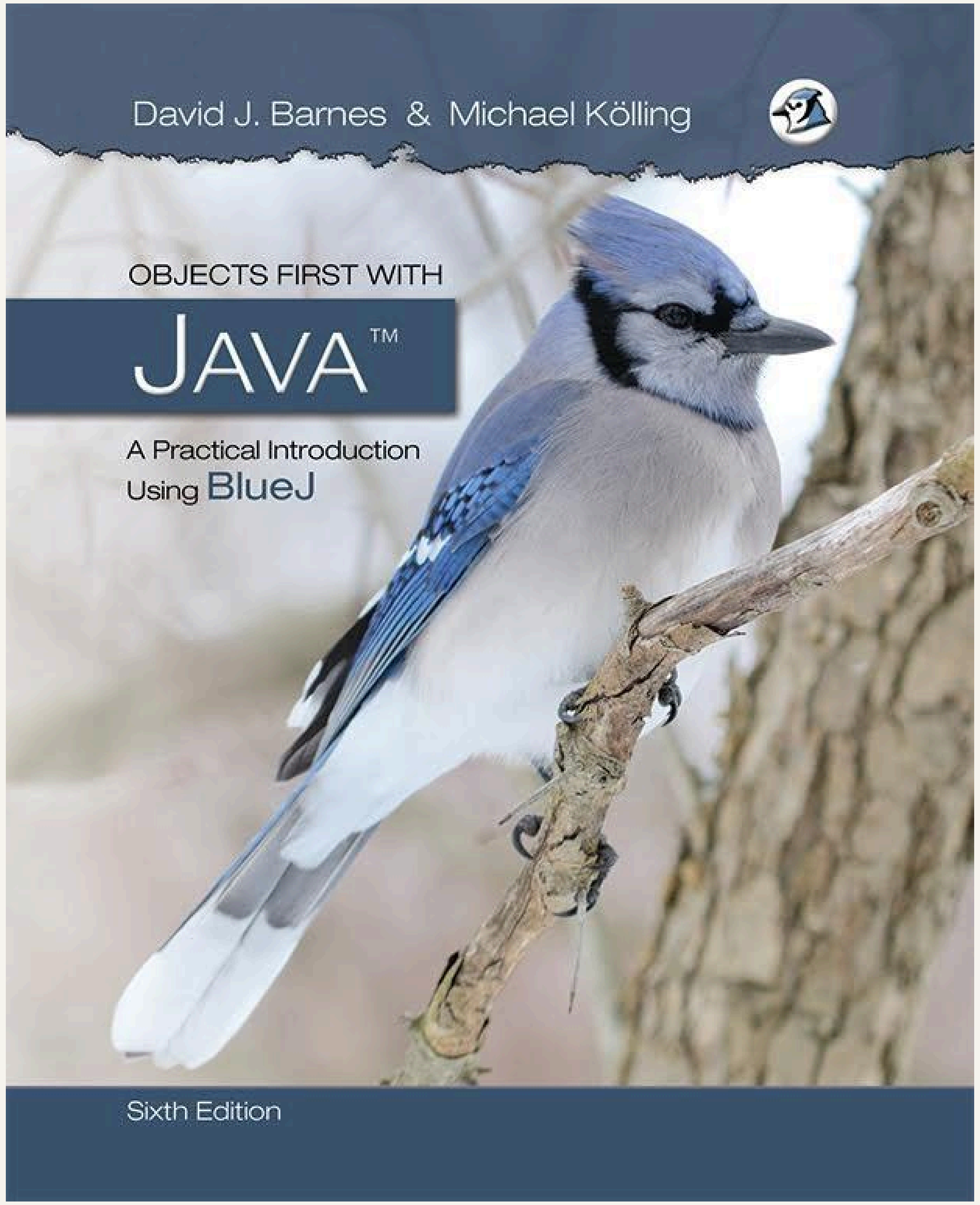
Thesis 30 ECTS

- Business Analytics and Data Science
- Creating Digital Systems
- Software Development Technology - foundational competences
- Software Development Technology - programming competence

- Frameworks and Architectures for the Web
- Functional Programming
- Symbolic AI and Optimization, MSc
- Mobile App Development

- Applied Algorithms
- Introduction to Security, MSc
- Big Data Management (technical)
- Data Mining and Machine Learning
- Distributed Systems
- Operating Systems and C
- Practical Concurrent and Parallel Programming
- Technical Interaction Design

Introductory Programming



This is BlueJ

SD: Ignore BlueJ, VS Code Only

They're just different interface for coding (IDE)

Learning

ITU Schedule

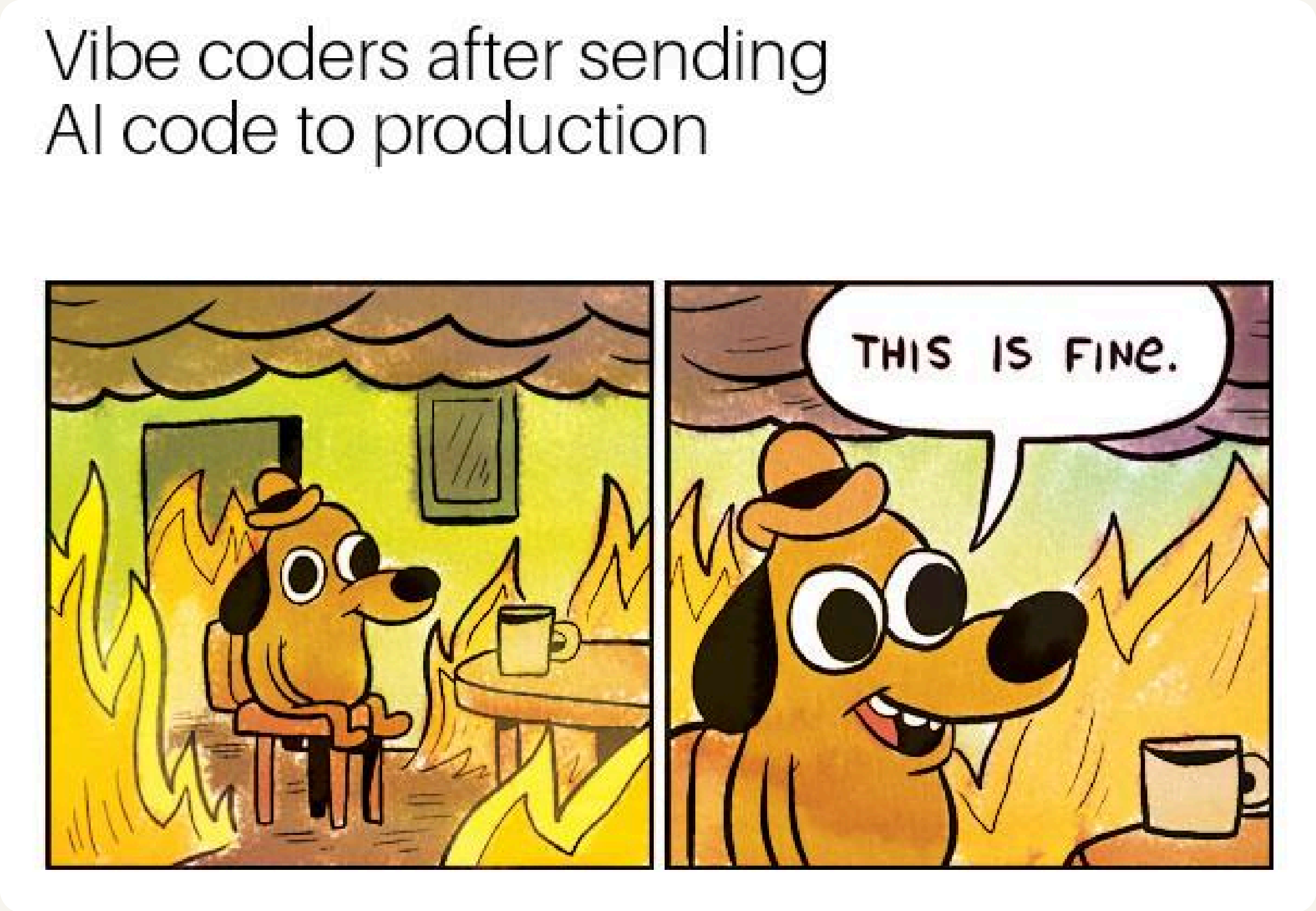
Passive

Lecture (5%)	Attending Lectures
Reading (10%)	Reading Materials
Audiovisual (20%)	Taking Notes
Demonstration (30%)	Attending Live Coding Sessions
Discussion Group (50%)	Joining Study Labs / Study Groups
Practice by Doing (75%)	Doing Exercises and Assignments
Teaching Others (90%)	Teaching each other / Being a TA

Active

Learning with AI

- Constantly ask "why?"
- Make a habit of reading code
- Take ownership and stay curious
- Treat AI as an essential skill
- Be honest with your progress



Programming Languages

- A **programming language** allows us (humans) to instruct computers what to do for us
- There are **many** (100.000s) of languages
  - Java, JavaScript, C, C++, C#, PHP, ...
- They have **different** characteristics
  - each has different pros & cons
  - the "best" language depends on your goals
  - Java can be used for small & large programs for different computers

Programming Languages

- Imperative Languages
  - Sequences of instructions
  - C, Pascal, ...
- Object-Oriented Languages
  - Manipulation of objects (abstractions from real world)
  - Java, C#, ...
- Functional Languages
  - "Mathematical" functions
  - ML, F#, ...
- Logical Languages
  - Mathematical Logic
  - Prolog, ...

Layers of Languages and Compiler

Compiler is the translator

Human Language

```
Keep adding to the count
as long as it's under 10.
```

</> Code

High-level Programming

```
while(x < 10){
    x = x + y;
}
```

Translate

Symbolic Machine Code

```
L: push x
   push y
   add
   pop x
   goto L
```

Machine Code

```
ff8a: push ffa8
      push 8a08
      add
      pop ffa8
      jump ff8a
```

Micro Code

```
push ffa8
noop p_reg %pc
noop push %r1
add store %pc
push 8a08
noop
```

Read

Hardware

```
100001100l0101
011101011010 01
0100010010 0100
1010101111111 1
000010101010 10
1101011001 0010
010101010101 10
110101010101 10
110111010001 ...
```

Coding Toolbox

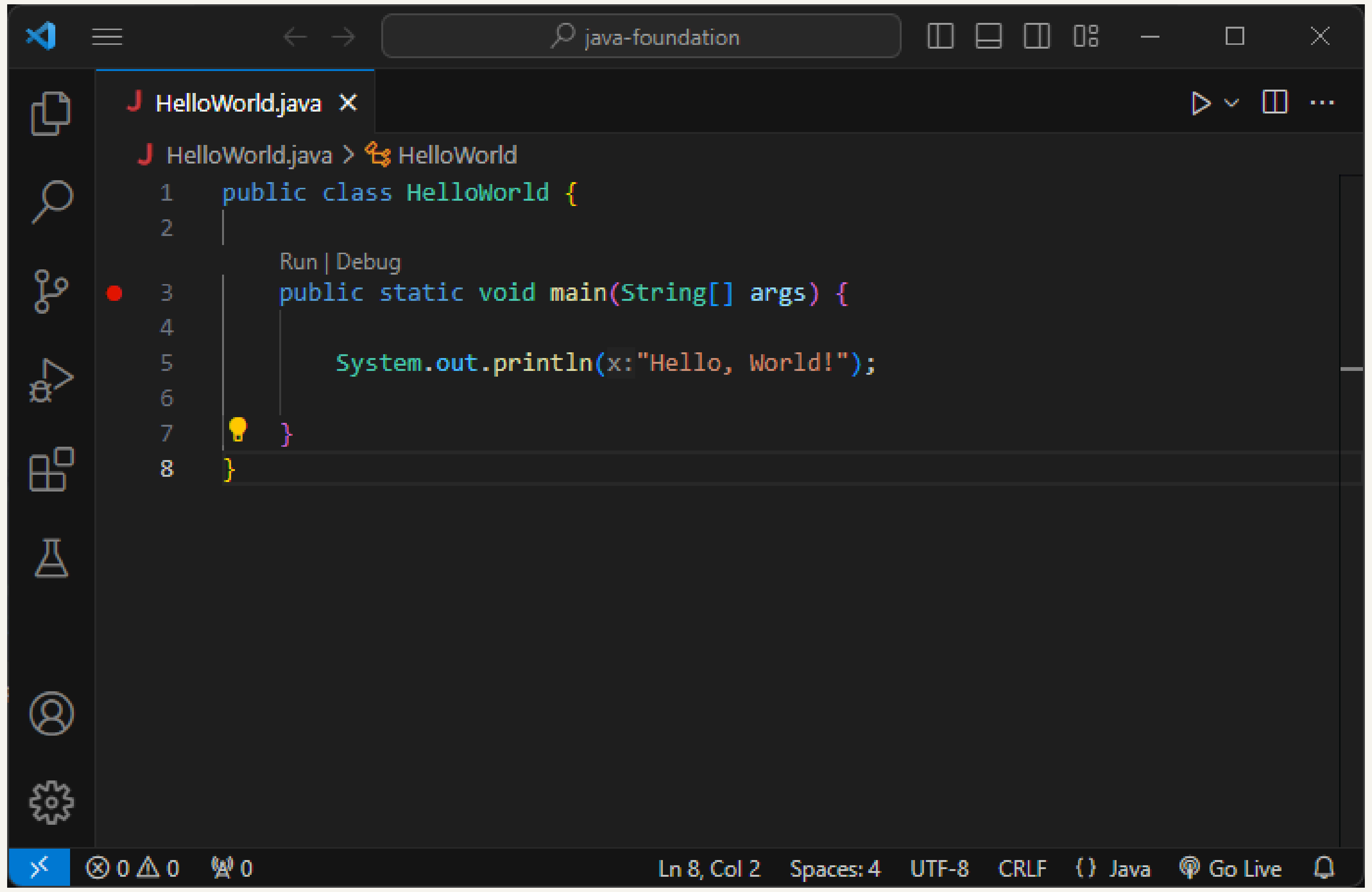
1 **JDK:** Compiles and runs your code.

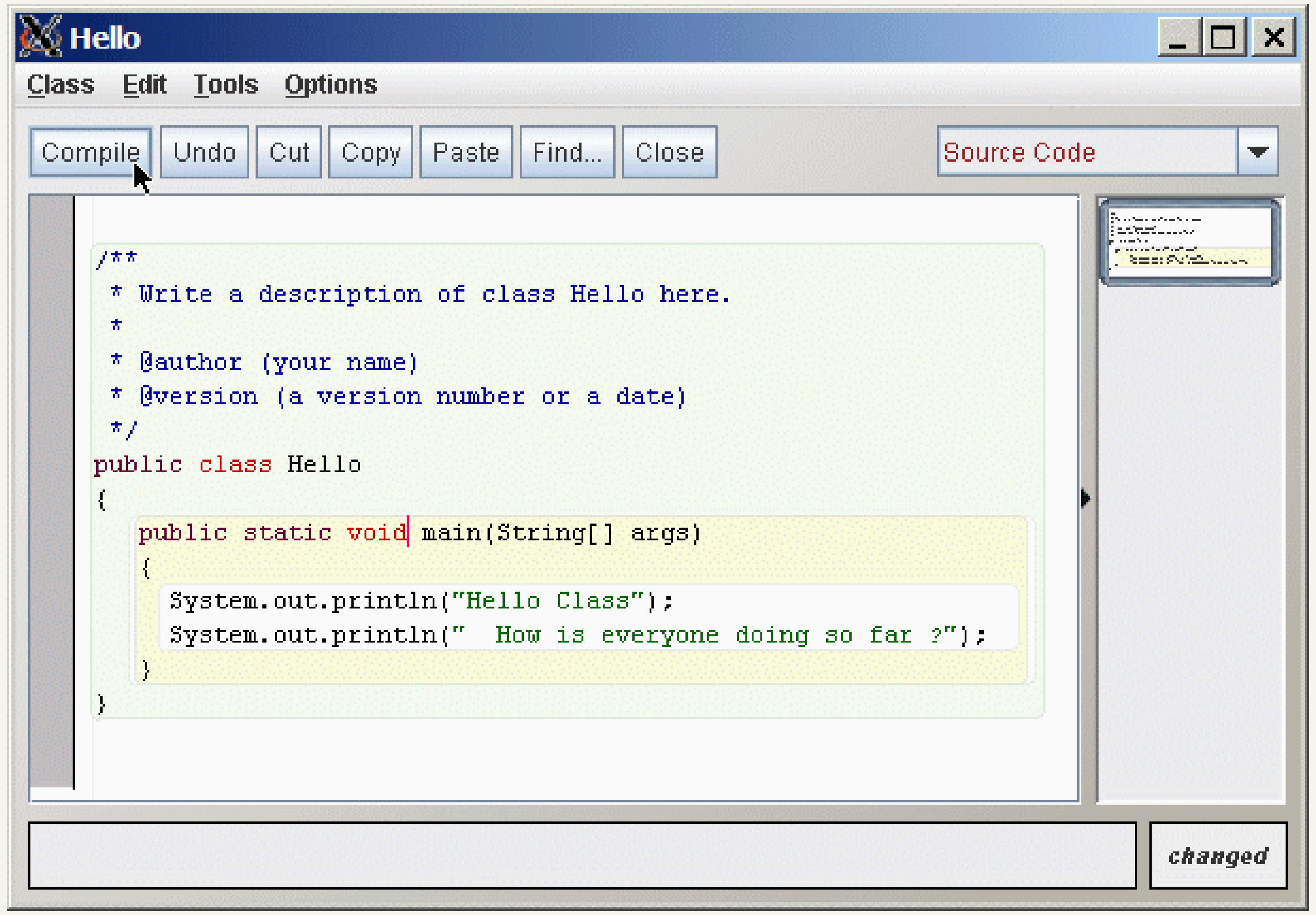
2 **IDE:** The code editor you will write in.

3 **VS Code Java Extension Pack:** autocomplete and error checking.

Coding Pack for Java

A single installer sets up the JDK, VS Code, and all required extensions.





BootCode - Online IDE Built for BootIT

Access the website via a laptop by...

Scanning the QR code

Entering the web address directly

BootCode - ITU BootIT

BootCode is the beginner-friendly Java coding platform for ITU BootIT, the pre-master computer science bootcamp. Write, run, and submit Java exercises directly in your browser.

cobblerscoders.tech

https://cobblerscoders.tech/

Clicking the link on the course platform

https://learnit.itu.dk/course/view.php?id=3026730

Your First Java Program (Probably!)

```java
public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

Your First Java Program (Probably!)

```java
public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

- **public class Main** is just the container for your code.
- **public static void main** is the front door of your app—it's where the program starts running.
- Don't worry about them for now. Simply put your code inside the main function's body.

Your First Java Program (Probably!)

```
public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

- **System.out.** A built-in toolbox provided by Java.
- **println()** Prints the text and moves to a new line.
- **print()** Prints the text without starting a new line.

- The actual message you want to print.
- Put it in the "here".

Exercise - Printing a Message


```java
public class Main {
    public static void main(String[] args) {
        System.out.println("Hello World!");
    }
}
```

Print a sentence to say hello to your new university.
For example: Hello ITU!

Run and test your code with BootCode,
then submit when you are confident with your solution.

Different Values

String: “test” between double quotes

```java
public static void main(String[] args) {
    System.out.println("Hello World!");
}
```

Integer: whole number

```java
public static void main(String[] args) {
    System.out.println(42);
}
```

Double: decimal number

```java
public static void main(String[] args) {
    System.out.println(3.14);
}
```

Exercise - Printing Different Values


```
System.out.println("Hello World!");
System.out.println(42);
System.out.println(3.14);
```

Your new friend is interested to know you better. Here are 3 questions from them, in order:

1. What's your name? — print it as **text**
2. Where do you live, in Danish zip code? — print it as a **whole number**
3. How long have you been here, in decimal years? — print it as a **decimal number**

Print your three answers in that exact order, **each on its own line.**

Note: if you don't feel like answering with your actual info, that's fine
— just print any 1. text, 2. whole number, 3. number with a decimal point, in that order.

Variables

Type Name Value

```java
int x = 42;
System.out.println(x);
```

Value: What is actually stored?

Name: Identifier used to refer to this box.

int **Type:** Shape of the box; What can be put in?

Types

int

```java
int x = 42;
System.out.println(x);
```

1, 2, 0, -2, -420, 1000000, 2147483647

String

```java
String s = "hi";
System.out.println(s);
```

"hi", "hello world", "@#$%&*(),.;", "14"

double

```java
double d = 3.14;
System.out.println(d);
```

1.5, 3.1415, -27.15, 1.0, 6.02E23

boolean

```java
boolean b = true;
System.out.println(b);
```

true and false

Exercise - Printing Different Values Using Variables


```
int x = 42;
System.out.println(x);
String s = "hi";
System.out.println(s);
```

```
double d = 3.14;
System.out.println(d);
boolean b = true;
System.out.println(b);
```

Another new friend is asking you the same 3 questions:

1. What's your name? — print it as **String**
2. Where do you live, in Danish zip code? — print it as a **int**
3. How long have you been here, in decimal years? — print it as a **double**

This time, **store each answer in a variable** first, so you can reuse it easily instead of repeating yourself. Then **print the three variables** in that exact order, each on its own line.

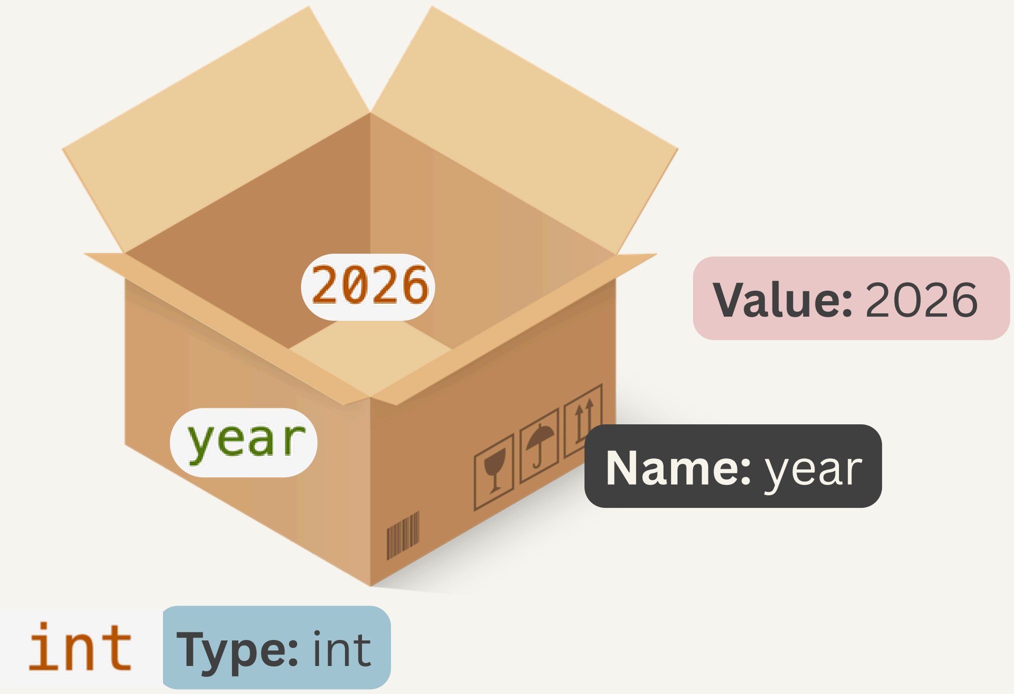
Note: if you don't feel like answering with your actual info, that's fine
— just print any 1. text, 2. whole number, 3. number with a decimal point, in that order.

Variable Assignment

Once a variable is declared and initialized, you can reassign it by using its name to set a new value of the same type.

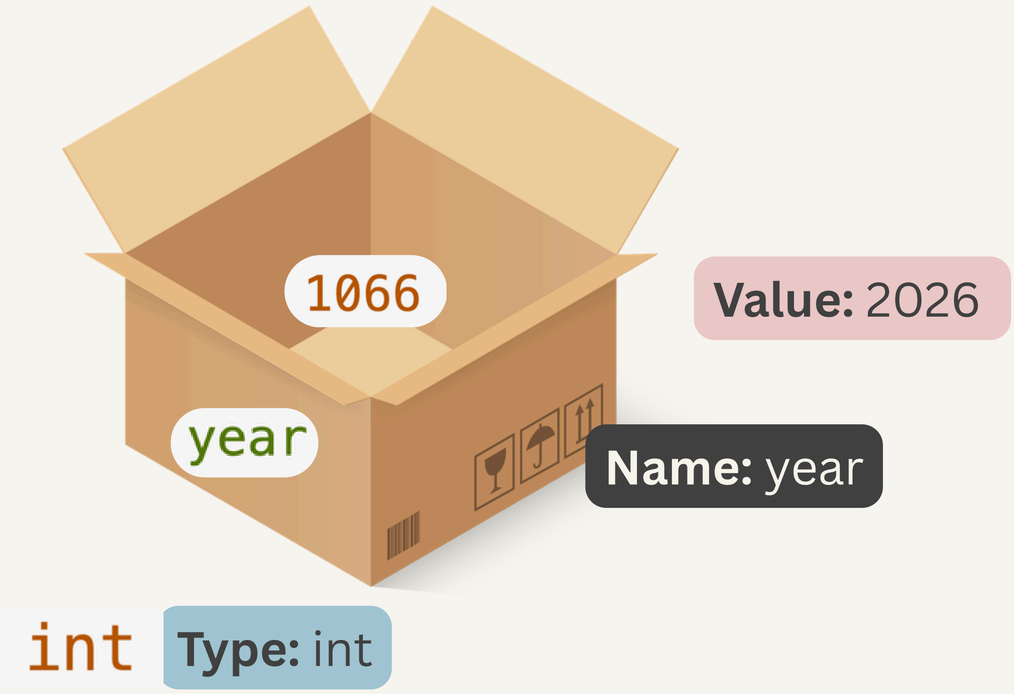
Initial Value ->

```java
int year = 2026;
System.out.print("The year is ");
System.out.println(year);
// The year is 2026
```



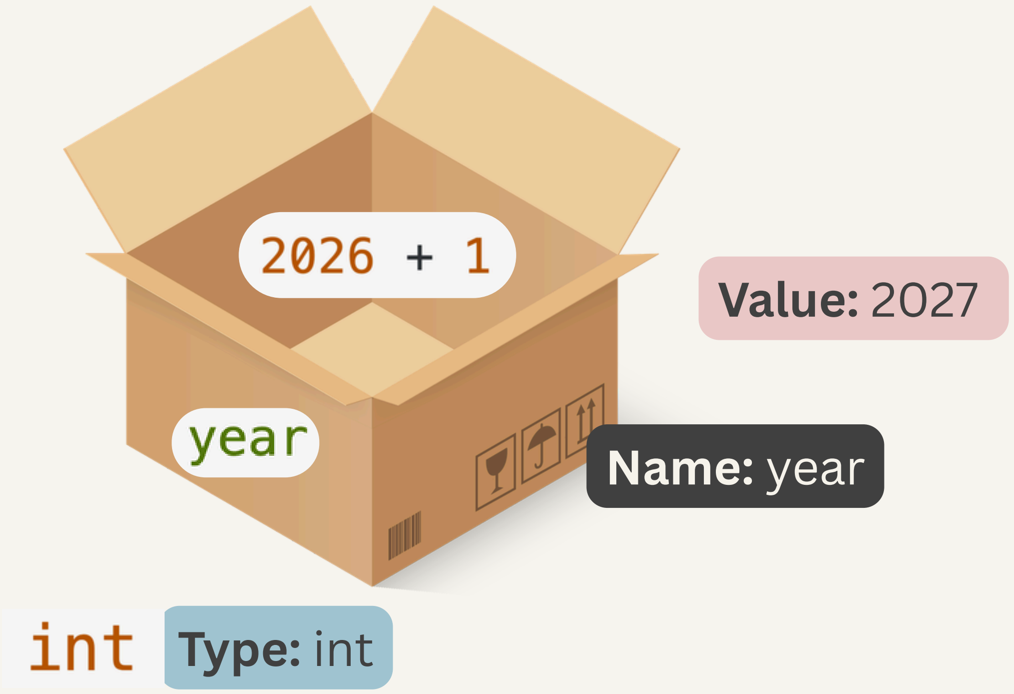
1. Using a new value

```java
year = 1066;
System.out.print("The year is now");
System.out.println(year);
// The year is now 1066
```



2. Calculating based on the old value

```java
year = year + 1;
System.out.print("The year is now ");
System.out.println(year);
// The year is now 2027
```



Exercise - Assign New Value to a Variable


```
int year = 2026;
System.out.print("The year is ");
System.out.println(year);
// The year is 2026

year = year + 1;
System.out.print("The year is now ");
System.out.println(year);
// The year is now 2027
```

Use one **int** variable called **age** (starting at 27) to print "ITU has been open for 27", then print the variable value after.

Then **update** the **SAME variable**, and print "Next year, ITU will have been open for 28".

Fun fact: ITU is the youngest university in Denmark, founded in 1999 — it turns 27 in 2026.

Operations on Numbers

Operators in int

```java
System.out.println(3 + 3);

int x = 3;
System.out.println(x + x);

x = 2;
System.out.println(x * x);

x = 6;
System.out.println(x / 3);
```

6

4

2

Operators in double

```java
System.out.println(2.2 + 2.2);

double x = 2.2;
System.out.println(x + x);

x = 2.5;
System.out.println(x * 3);

x = 5.0;
System.out.println(x / 2);
```

4.4

7.5

2.5

Anatomy of Java Code

- **Statement**: A complete unit of execution (usually ending with a semicolon ;).
- **Expression**: A code snippet that evaluates to a single value.
- **Literal Value**: A fixed value written directly in the source code.
- **Variable**: A named storage location of a specific type that holds a value.

Variable

Literal Value

1

2

```
int x = 2;
System.out.println((4 + (x * x) + (4 - x)) / x);
```

3

4

Expressions

Statement

Exercise - Use Numeric Operators


```java
System.out.println(3 + 3);   // 6

int x = 2;
System.out.println(x * x);   // 4

int y = 6;
System.out.println(y / 3);   // 2
```

Fun fact: According to ITU's 2025 figures, about 23.4% of students are international. The Master in Software Design program admits around 130 students per year.

Write code to **calculate and print** the estimated number of international students in this program.

Operations on Strings

+ is String Concatenation Operatort

```
String hi = "Hello ";
String world = "World!";
String greet = hi + world;

System.out.println(greet);
// Hello World!
```

It also works between Strings and numbers:

```
int year = 2026;
System.out.println("The year is " + year);
```

Exercise - Concatenate Strings

```java
String hi = "Hello ";
String world = "World!";
String greet = hi + world;

System.out.println(greet);
 // Hello World!
```

```java
int year = 2026;
System.out.println("The year is " + year);
```

Ask the person sitting next to you for their name, then modify the code to greet them personally: Hello my friend, {Name}!

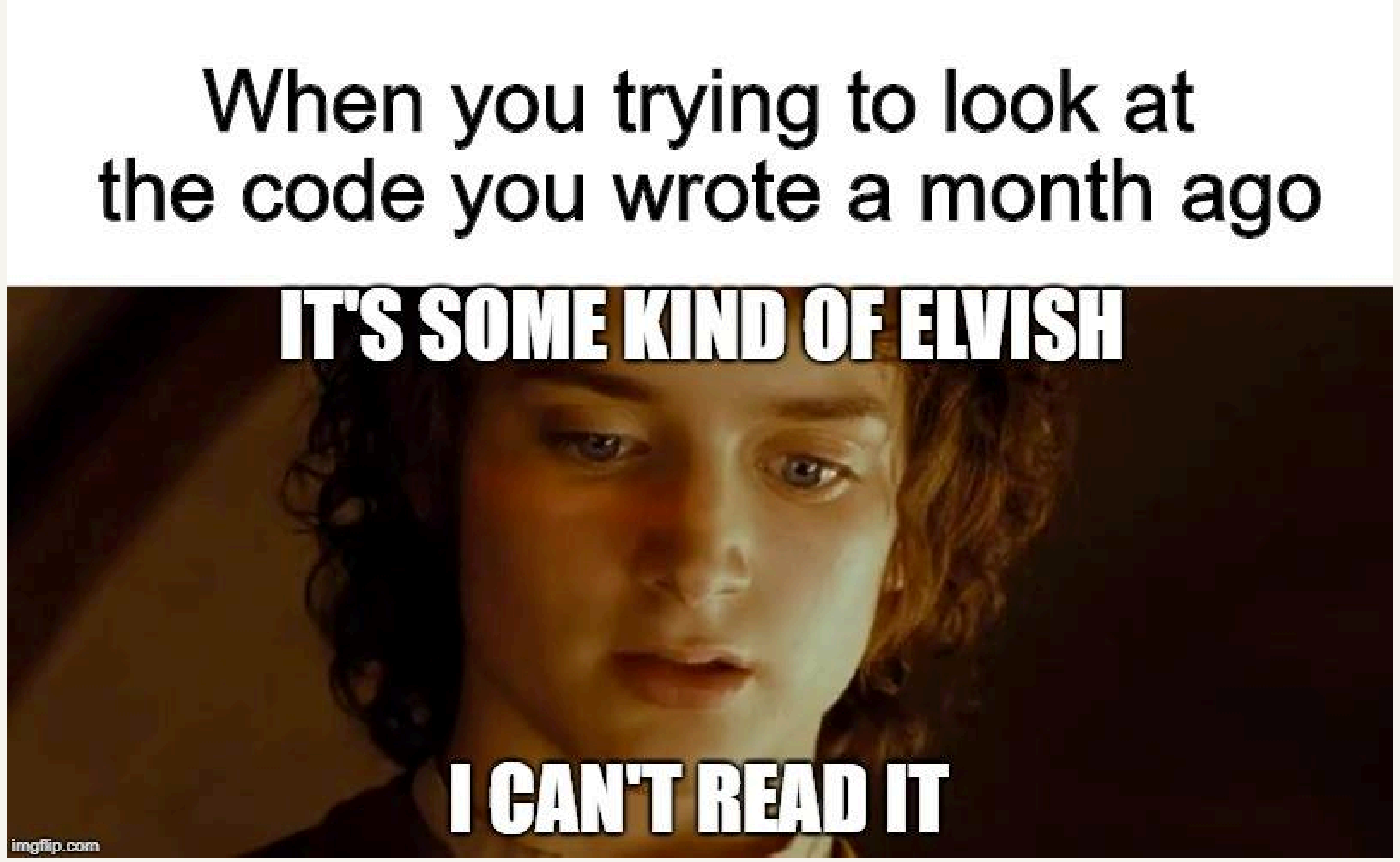
Declare a variable to hold the name, and print the **concatenated sentence**.

Comments

Text in the source code meant for **humans** (and AI) to **read**, which is completely **ignored by** the **compiler** when executed.

Why use them?

- Easier to understand / read the code
- Help others understand the code
- Help future you understand the code



Comments

- Use **//** for single line comment

```java
public static void main(String[] args) {
    // Hi, I am a comment!
    System.out.println("Hello World!");
}
```

- Use **/* text in between */** for multiple lines comment

```java
public static void main(String[] args) {
    /* I am in fact also a comment,
   but I can span multiple lines
   :-) */
   System.out.println("Hello World!");
}
```

Exercise - Currency Exchange

During the break you meet a friend at ITU's own café, Cafe Analog.
A coffee costs 20 dkk. Your friend says that is cheap — but you want to see it in euro.
This code converts the other way, from euro to kroner:

```
public class Main {
    public static void main(String[] args) {
        int eur = 100;
        double dkk = eur * 7.45;
        System.out.println(eur + " euro corresponds to " + dkk + " kr.");
    }
}
```

Modify the code so it **converts** the opposite way: **from kroner to euro**.
For a 20 dkk coffee, print: "20 dkk corresponds to {eur} euro."

Note: All the decimals are fine, that is just how doubles print.

Methods

Methods let us **reuse code** and give a snippet a clear **responsibility**.

This does exactly the same as the previous exercise, wrapped in a method:

```java
public class Main {
    static void dkkToEur() {
        double dkk = 100;
        double eur = dkk / 7.45;
        System.out.println(dkk + " kr corresponds to " + eur + " euro");
    }

    public static void main(String[] args) {
        dkkToEur();
    }
}
```

Exercise - Declare a Method


```java
public class Main {
    static void dkkToEur() {
        double dkk = 100;
        double eur = dkk / 7.45;
        System.out.println(dkk + " kr corresponds to " + eur + " euro");
    }

    public static void main(String[] args) {
        dkkToEur();
    }
}
```

Declare a new **function eurToDkk()** that converts 100 euro to dkk and **prints** "{eur} euro corresponds to {dkk} kr".

Call it from main, after dkkToEur().

Methods with Parameters

But the previous method always converts 100 kr.

With a **parameter**, the same method works for any value:

```java
public class Main {
    static void dkkToEur(double dkk) {
        double dkk = 100;
        double eur = dkk / 7.45;
        System.out.println(dkk + " kr corresponds to " + eur + " euro");
    }

    public static void main(String[] args) {
        dkkToEur();
        dkkToEur(100);
        dkkToEur(20);
    }
}
```

Sequence of Function

When a method is called, the execution jumps to that method.
It returns to the calling line after finishes executing .

```
class Valuta {
2 → static void dkk2eur(double dkk) {
        double eur = dkk / 7.45;
        System.out.println(dkk + " kr corresponds to " + eur + " euro");
    }
1 → public static void main(String[] args) {
        dkk2eur(100); → 2
        dkk2eur(20); → 2
    }
}
```

Within a method, the program executes line by line from top to bottom.

The main method is the entry point of a Java program, where execution always begins.

main → dkk2eur(100) → lines in dkk2eur → dkk2eur(20) → lines in dkk2eur

Exercise - Declare a Method with Parameters

```
public class Main {
    static void dkkToEur(double dkk) {
        double eur = dkk / 7.45;
        System.out.println(dkk + " kr corresponds to " + eur + " euro");
    }

    public static void main(String[] args) {
        dkkToEur(100);
        dkkToEur(20);
    }
}
```

The previous dkkToEur and eurToDkk always convert a fixed 100.

Rewrite both so they take a parameter instead:

- dkkToEur(double dkk) — prints "{dkk} kr corresponds to {eur} euro"
- eurToDkk(double eur) — prints "{eur} euro corresponds to {dkk} kr"

Call dkkToEur(100) and eurToDkk(100) from main — the output should match the previous exercise, but now the same functions work for any amount

Call Stack

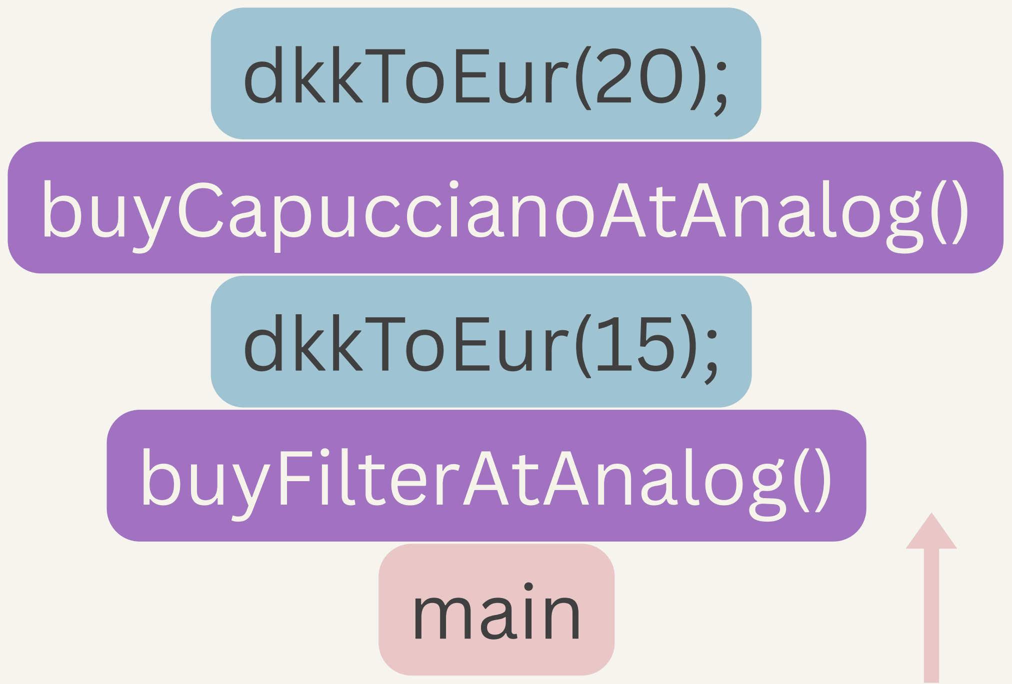
- A method with several parameters can print many different values the same way.
- A method isn't only called from main — it can call another method too.
- A high-level method only knows what to do, without caring how it is done.

```java
public class Main {
    static void dkkToEur(double dkk) {
        double eur = dkk / 7.45;
        System.out.println(dkk + " kr corresponds to " + eur + " euro");
    }

    static void buyFilterAtAnalog() {
        System.out.println("A cup of filter coffee cost " + 15 + " dkk");
        dkkToEur(15);
    }

    static void buyCapuccianoAtAnalog() {
        System.out.println("A cup of capucciano cost " + 20 + " dkk");
        dkkToEur(20);
    }

    public static void main(String[] args) {
        buyFilterAtAnalog();
        buyCapuccianoAtAnalog();
    }
}
```



Exercise - Methods Call Stack

At ITU every course is worth ECTS points, and a full semester adds up to 30 ECTS. You are starting Software Design. **Write printCourse(String name, double ects, int semester)**, to print one course's line.

Then **build one layer on top of it**: a method per semester that calls printCourse for each of that semester's courses.

- printFirstSemester() calls printCourse(): Introductory Programming (15 ECTS), Discrete Mathematics (7.5 ECTS), Software Engineering (7.5 ECTS)
- printSecondSemester() calls printCourse(): Introduction to Database System (7.5 ECTS), Algorithm and Data Structures (7.5 ECTS)

main calls **printFirstSemester()** and **printSecondSemester().**